

Teaching Exam

Subject – Computer Science

**Topic – Computer Organization &
Architecture**



Lecture No.- 05

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Topics to be Covered

Topic



- 1:- Demultiplexer- 1x2,1x4,1x8,1x16 Demux
- 2:- Create Higher Order Demux from Lower Order Demux
- 3:- Encoder
- 4:- Types of Encoder
- 5:- BCD , Hexadecimal , Octal all this to Binary Conversion using Encoder
- 6:- Priority Encoder
- 7:- Advantage, Application , Disadvantage & Conclusion On Encoder.



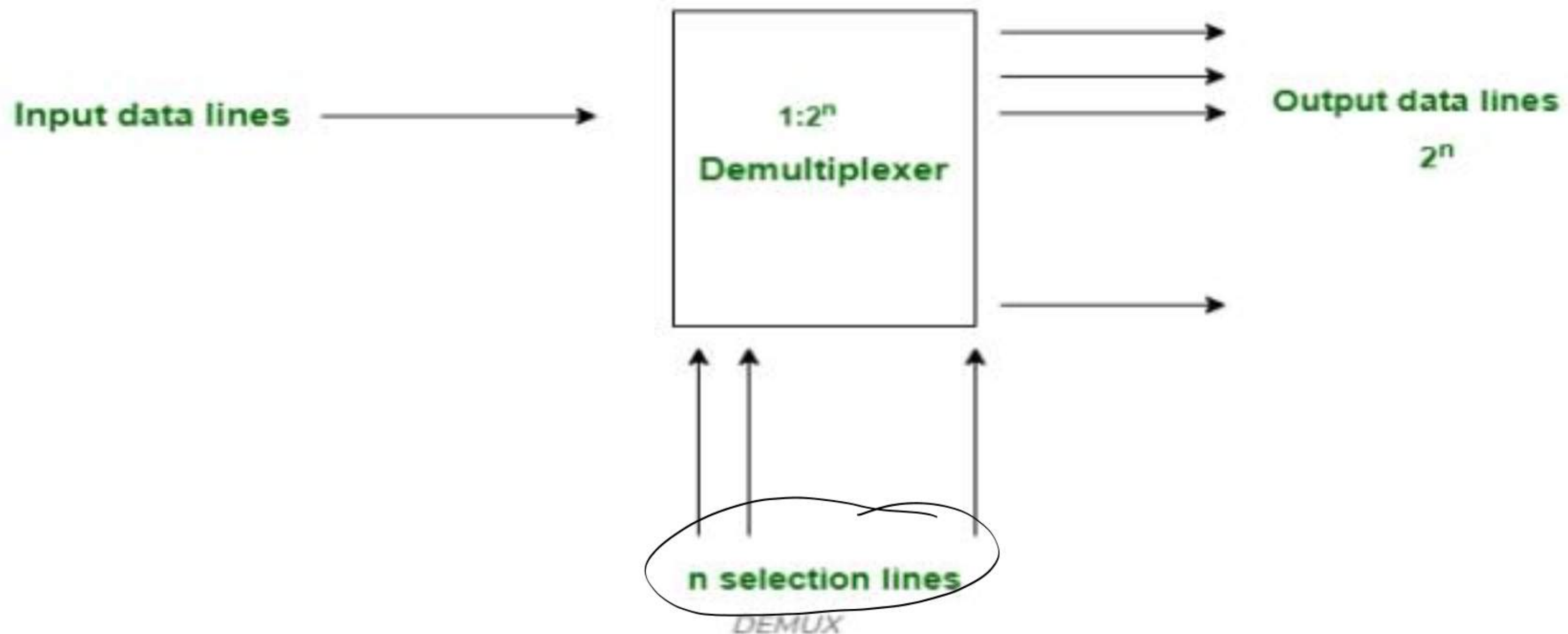
What is a DEMUX?

The DEMUX is a digital information processor. It takes input from one source and also converts the data to transmit towards various sources. The demultiplexer has one data input line. The demultiplexer has several control lines (also known as select lines). These lines determine to which output the input data should be sent. The number of control lines determines the number of output lines.



General Block Diagram Of A DEMUX

Here is the basic block diagram of a DEMUX as mentioned below.

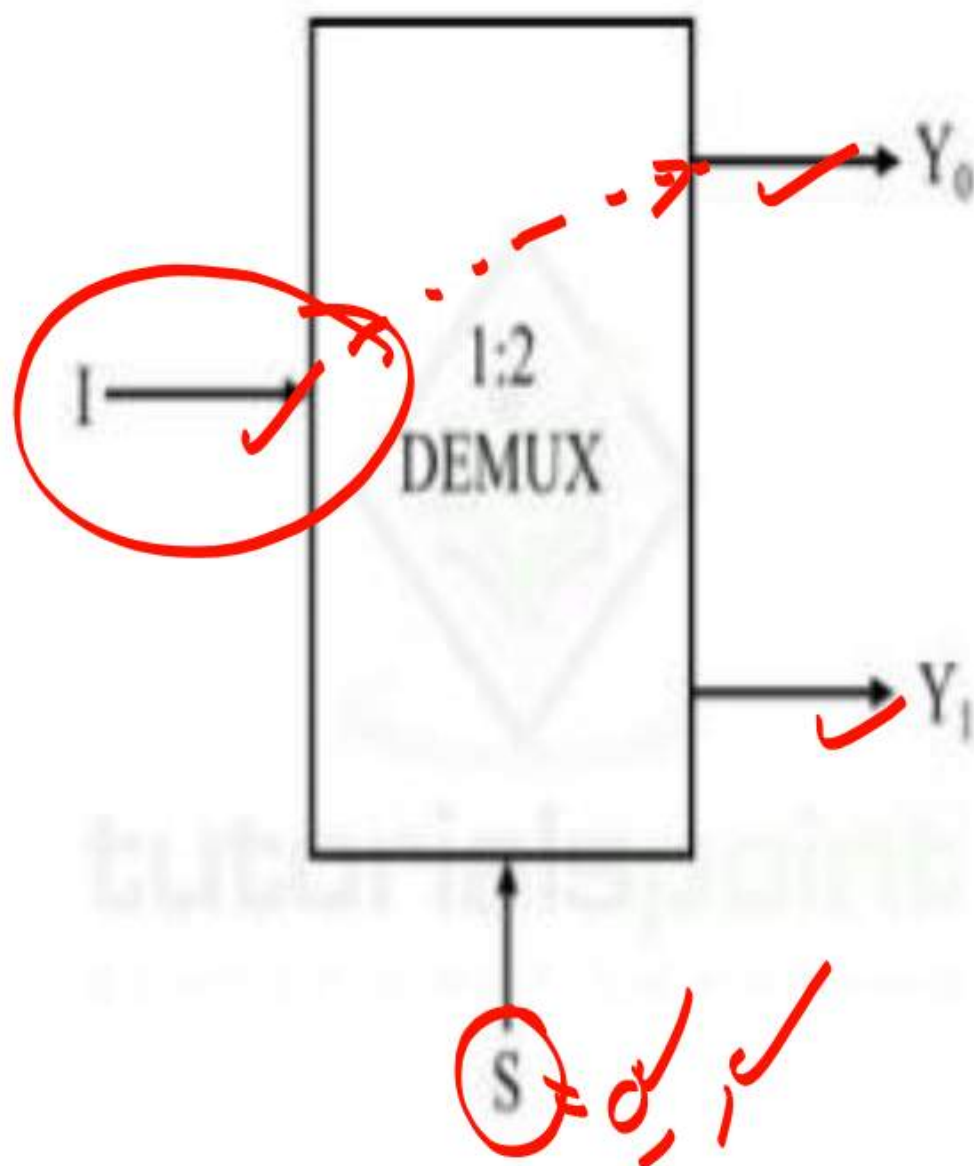




1x2 Demultiplexer

The operation of the 1x2 Demultiplexer can be analyzed with the help of its function table given below.

The functional block diagram of a 1x2 Demultiplexer is shown in



Select Line	Outputs	
S	Y ₁	Y ₀
0	0	1
1	1	0

From this function table of 1x2 Demultiplexer, we can directly derive the Boolean expression for each output as follow.

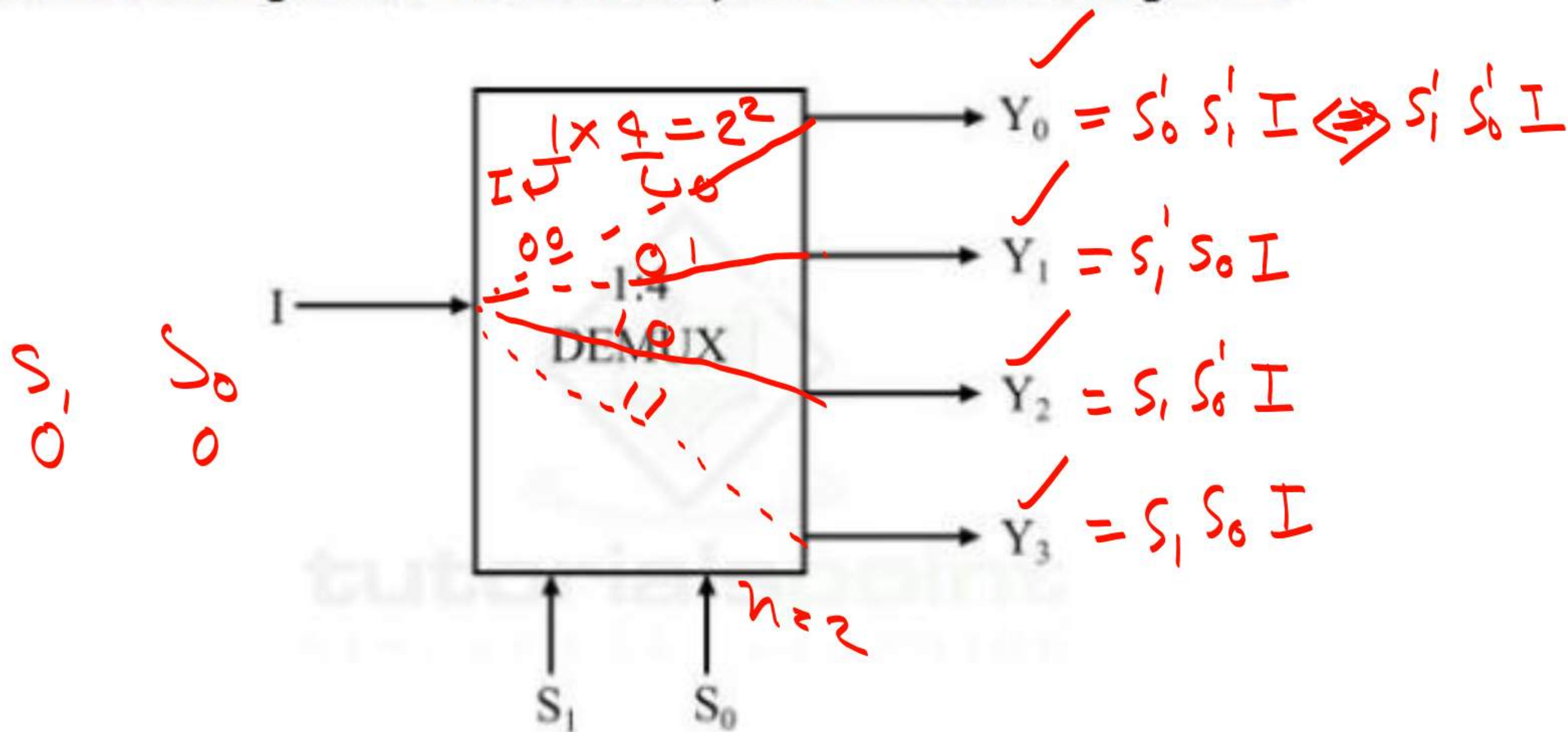
And,

$$\begin{aligned} Y_0 &= \bar{S}I \\ Y_1 &= SI \end{aligned} \quad \left. \begin{array}{l} \text{Individual} \end{array} \right\} \Rightarrow \bar{S}I, SI$$



1x4 Demultiplexer

The functional block diagram of 1x4 Demultiplexer is shown in Figure-3.





Topic : 1x4 Dmux



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The operation of the 1×4 Demultiplexer can be understood with the help of its function table given below.

Select Line		Outputs			
S_1	S_0	Y_3	Y_2	Y_1	Y_0
0	0	0	0	0	I
0	1	0	0	I	0
1	0	0	I	0	0
1	1	I	0	0	0

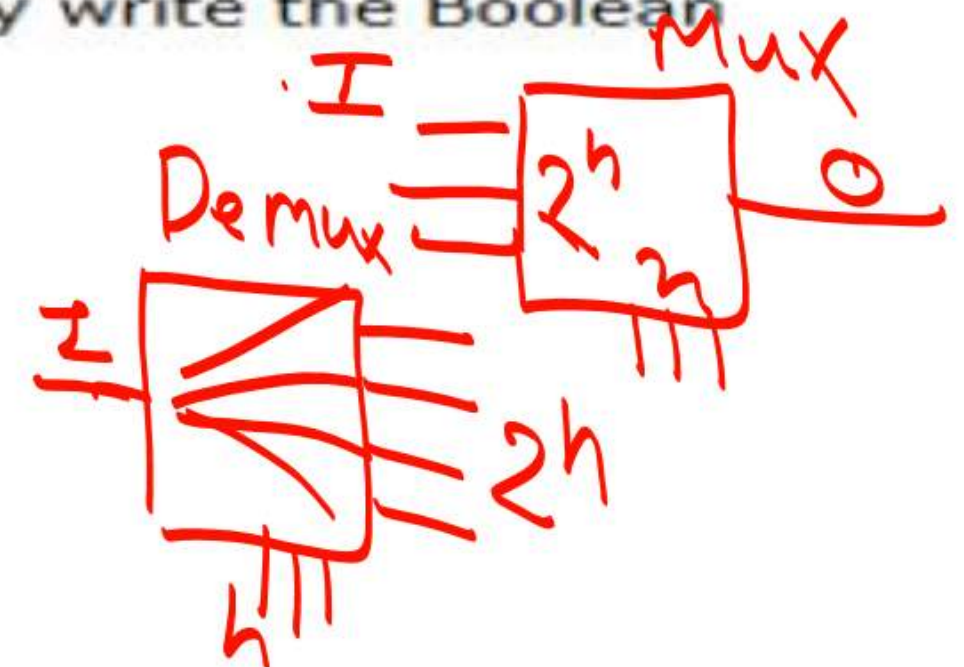
From this truth table of 1×4 Demultiplexer, we can directly write the Boolean expression for each output as follow.

$$Y_0 = \bar{S}_1 \bar{S}_0 I$$

$$Y_1 = \bar{S}_1 S_0 I$$

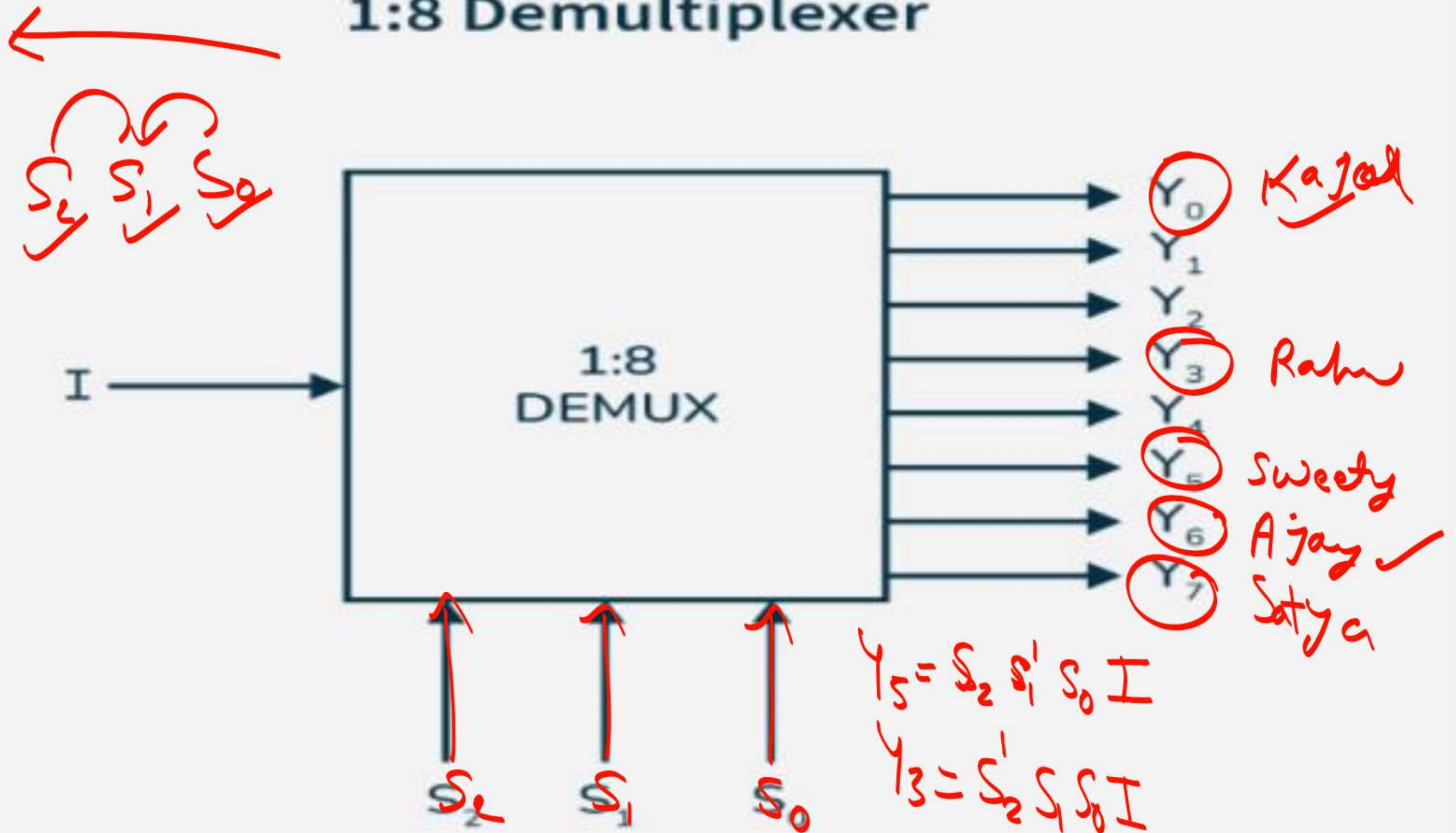
$$Y_2 = S_1 \bar{S}_0 I$$

$$Y_3 = S_1 S_0 I$$





1:8 Demultiplexer





Topic : 1x8 Dmux



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Let the 1×8 Demultiplexer has one input I , three selection lines s_2 , s_1 & s_0 and outputs Y_7 to Y_0 . The **Truth table** of 1×8 Demultiplexer is shown below.

Selection Inputs			Outputs							
s_2	s_1	s_0	Y_7	Y_6	Y_5	Y_4	Y_3	Y_2	Y_1	Y_0
0	0	0	0	0	0	0	0	0	0	I
0	0	1	0	0	0	0	0	0	I	0
0	1	0	0	0	0	0	0	I	0	0
0	1	1	0	0	0	0	I	0	0	0
1	0	0	0	0	0	I	0	0	0	0
1	0	1	0	0	I	0	0	0	0	0
1	1	0	0	I	0	0	0	0	0	0
1	1	1	I	0	0	0	0	0	0	0



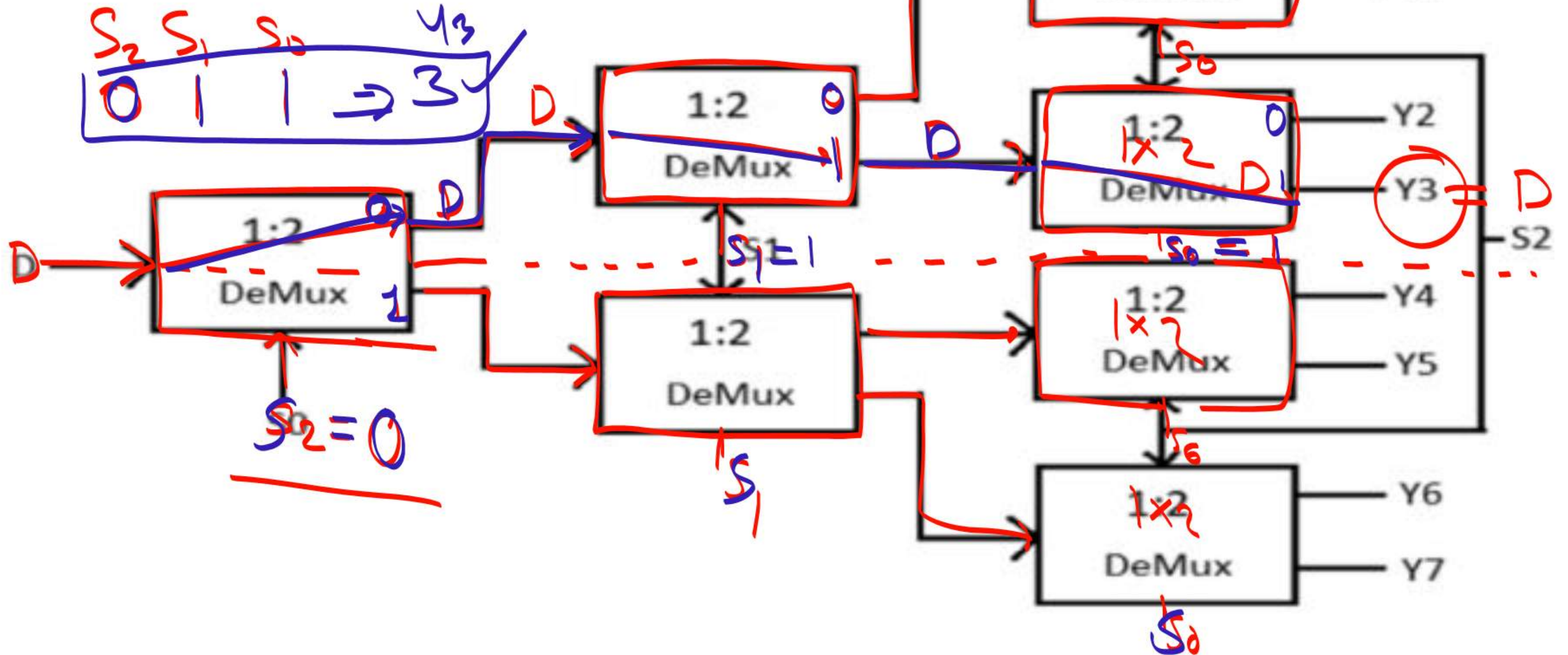
Topic : 1x8 Dmux using 1x2 Demux



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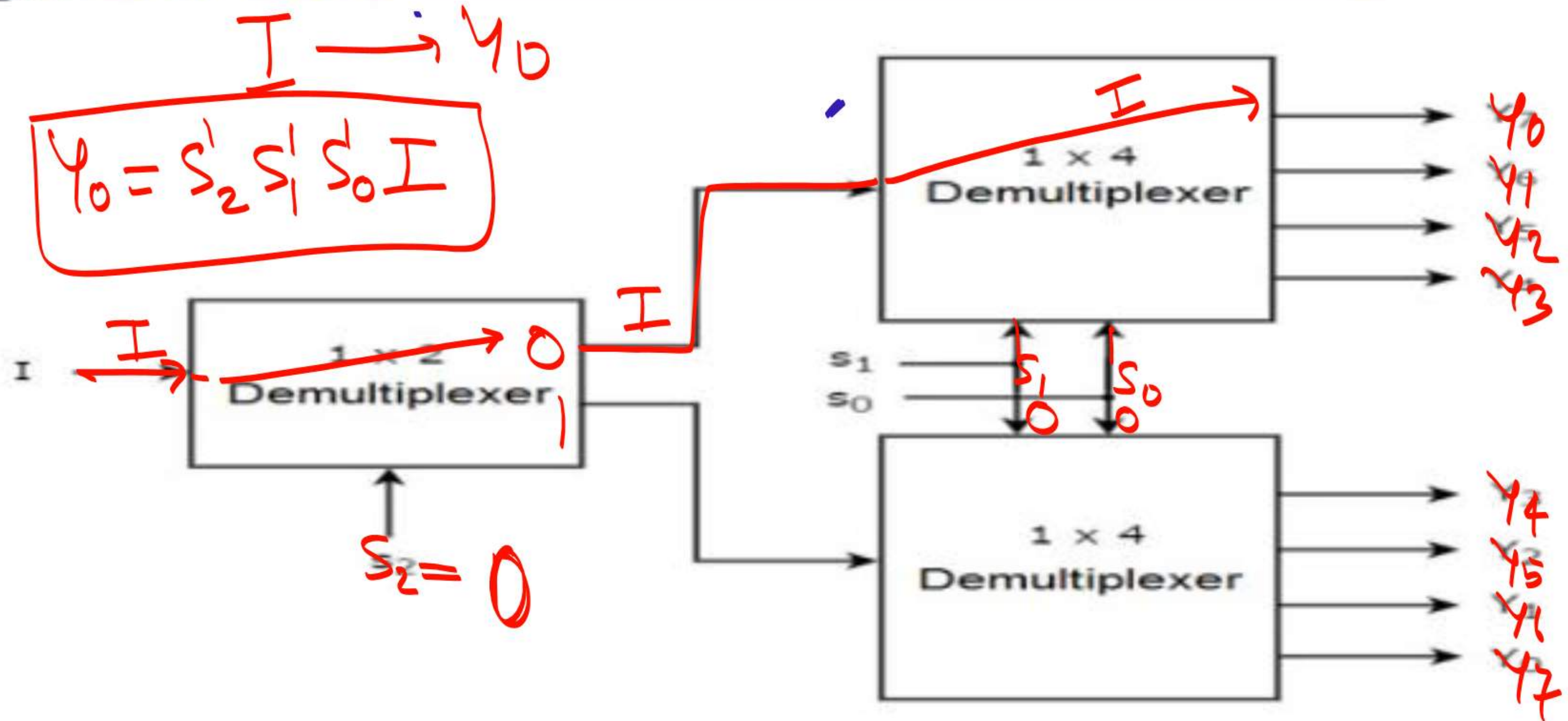
$$Y_3 = S_2^1 S_1 S_0 D$$

$$\frac{0}{2} = (4), \frac{4}{2} = (2), \frac{2}{2} = (1)$$





Topic : 1x8 Dmux using 1x4 Demux and 1x2 Demux- Special Case



8/4=2 , 2/4= 0.5 That is why 1x8 does not create by only 1x4 and it will take help by 1x2 demux as well.



1 X 16 Demux



Create it own by you, It is Home-Work

S_3 S_2 S_1 S_0
1 1 0 0

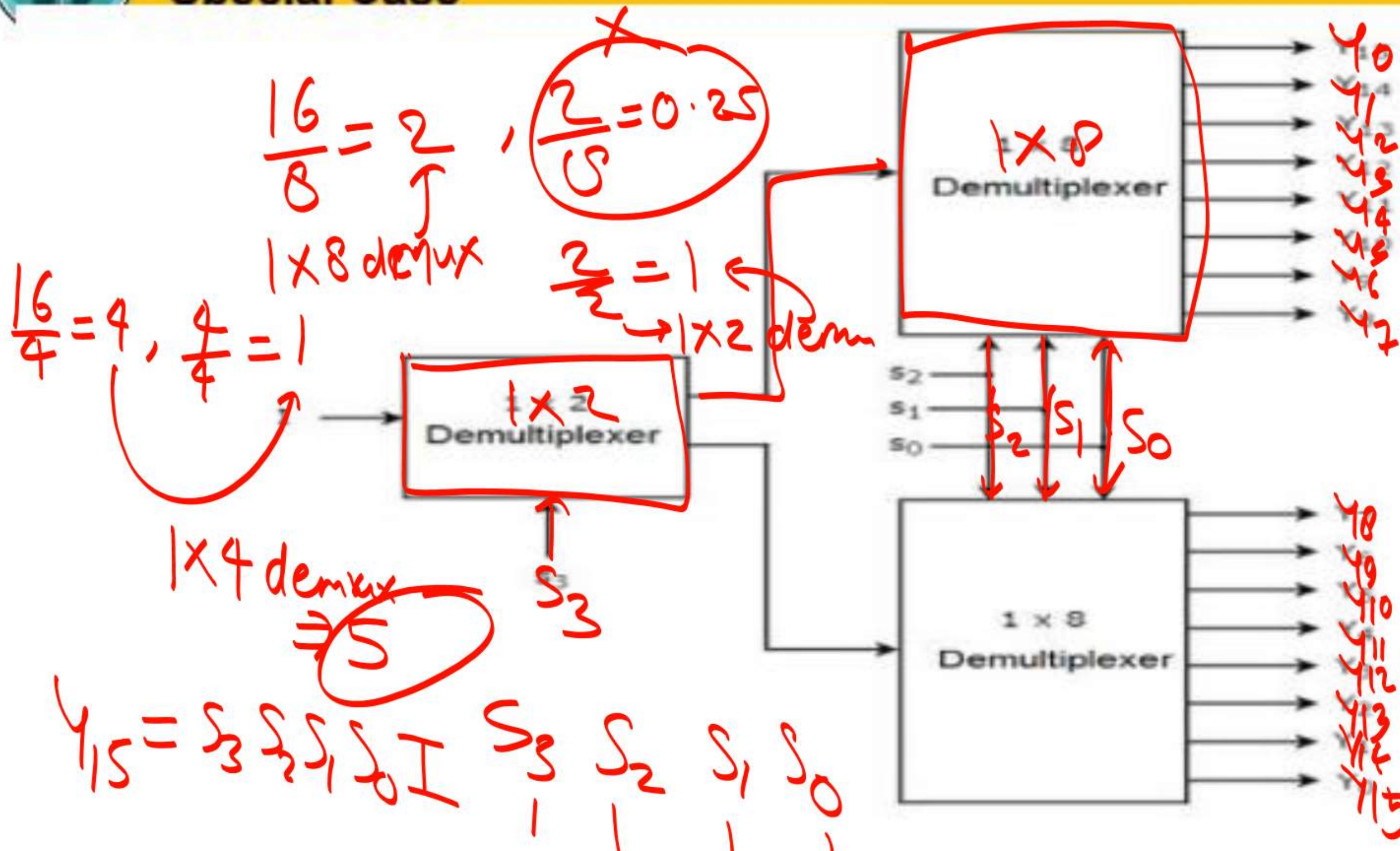
$$Y_{12} = S_3 S_2 S_1' S_0' I$$



Topic : 1x16 Dmux using 1x8 Demux and 1x2 Demux- Special Case



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$\rightarrow A7a0 = s_3 s_2 s_1 s_0 I$

$y_{11} = \text{Kajal}$

$y_{15} \rightarrow \text{Stylo}$

$y_{13} \rightarrow \text{Sweety}$

$y_{13} = s_3 s_2 s_1 s_0 I$

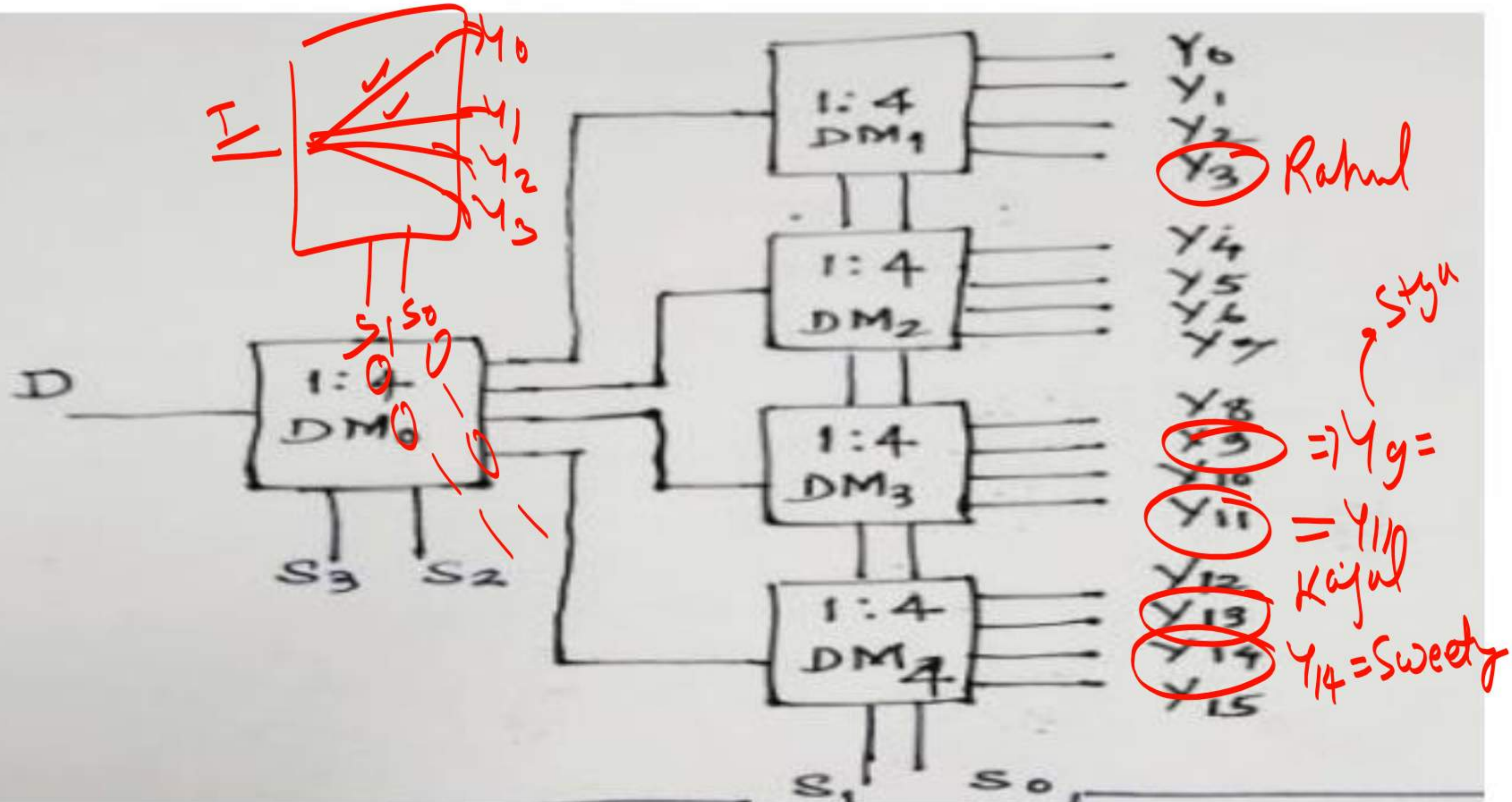
16/8=2 , 2/16= 0.125 That is why 1x16 does not create by only 1x8 and it will take help by 1x2 demux as well.



Topic : 1x16 Dmux using 1x4 Demux



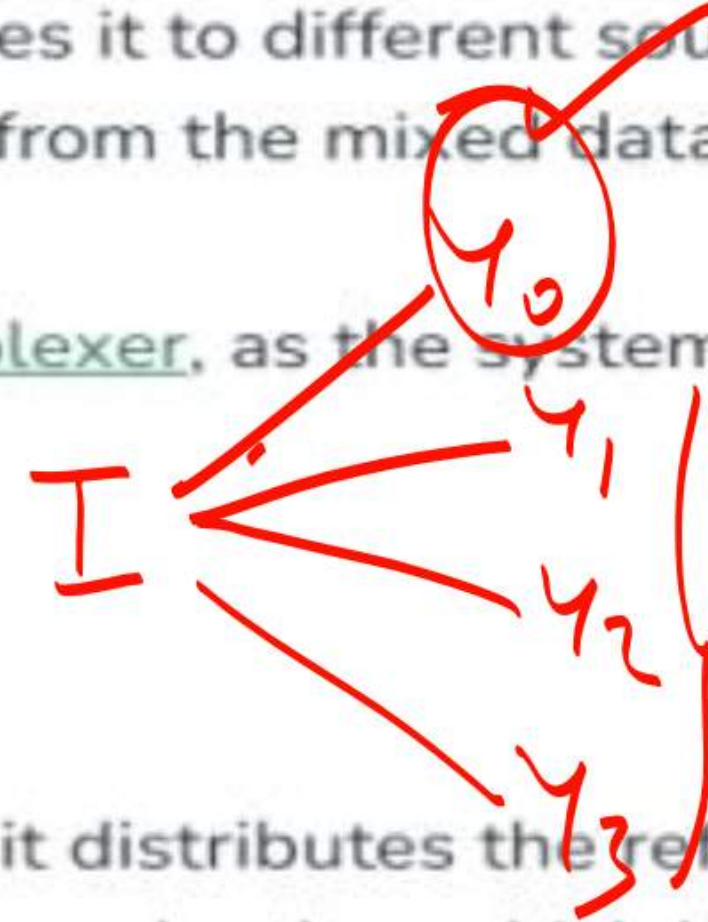
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Advantages of the DEMUX

- The DEMUX increases the efficiency of the particular communication system as it takes data from a specific input source and distributes it to different sources.
- The DEMUX helps to separate the different signals from the mixed data sources. Then it distributes these data to different sources.
- DEMUX can decode the signal outputs of the multiplexer, as the system works in a reverse way of the MUX.



Disadvantages of the DEMUX

- The DEMUX may cause a wastage of bandwidth as it distributes the refined data in different channels. These channels can overlap with each other which leads to the loss of signal.
- The DEMUX may cause problems in the synchronization of signals. The data channels can overlap with each other which leads to the delay in the whole process.



Applications of DEMUX

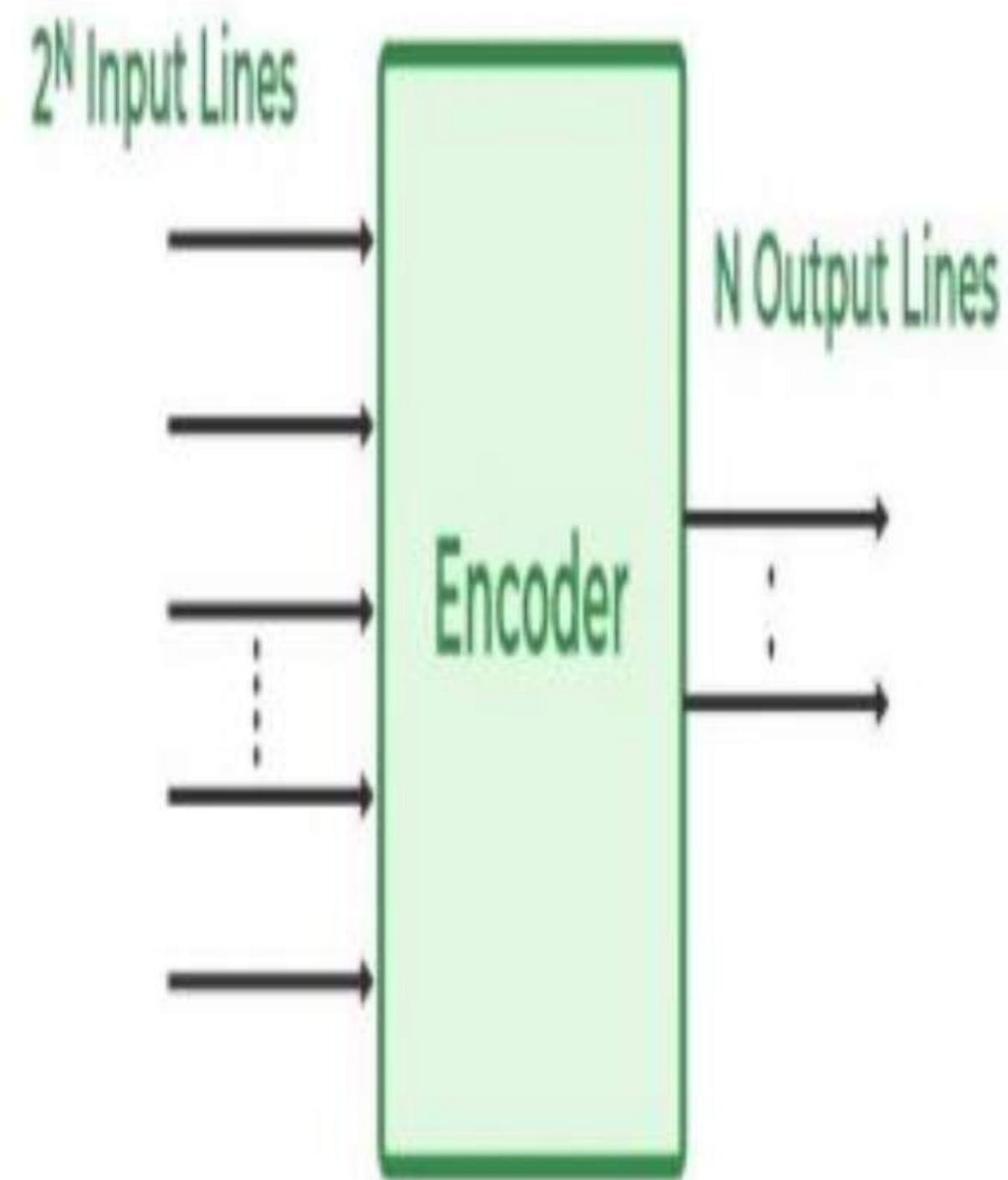
- DEMUXs are majorly used to design the ALU circuits and parallel data segments.
- The DEMUXs convert the output of the MUX into the actual input. On the receiver end, the DEMUX can be used to verify the original form of the data to carry out the entire communication process.
- DEMUX helps to save the output. The output generally saved to the ALU. The output will be saved in the registers and the various storage units of the system.
- Each DEMUX has a connection with multiple registers which helps to store the processed data into it.
- The counterpart of the DEMUX regulates the data signal at the output stage of the DEMUX operation. These data can be retrieved later to read out parallelly.
- In the audio or video system, the DEMUX distributes them in different channels. In broadcasting, the DEMUX separates the composite signal into an individual one.



What is Encoder?

An Encoder is a **combinational circuit** that performs the reverse operation of a [Decoder](#). It has a maximum of 2^n input lines and 'n' output lines, hence it encodes the information from 2^n inputs into an n-bit code. It will produce a binary code equivalent to the input, which is active High. Therefore, the encoder encodes 2^n input lines with 'n' bits.

The basic principle of an encoder is to assign a unique binary code to each possible input. For example, a 2-to-4 line encoder has 2 input lines and 4 output lines and assigns a unique 4-bit binary code to each of the $2^2 = 4$ possible input combinations. The output of an encoder is usually active low, meaning that only one output is active (low) at any given time, and the remaining outputs are inactive (high). The active low output is selected based on the binary code assigned to the active input.



Types of Encoders

There are different types of Encoders which are mentioned below.

- 4 to 2 Encoder
- Octal to Binary Encoder (8 to 3 Encoder)
- Decimal to BCD Encoder
- Priority Encoder



The 4 to 2 Encoder consists of four inputs Y3, Y2, Y1 & Y0, and two outputs A1 & A0. At any time, only one of these 4 inputs can be '1' in order to get the respective binary code at the output. The figure below shows the logic symbol of the 4 to 2 encoder.





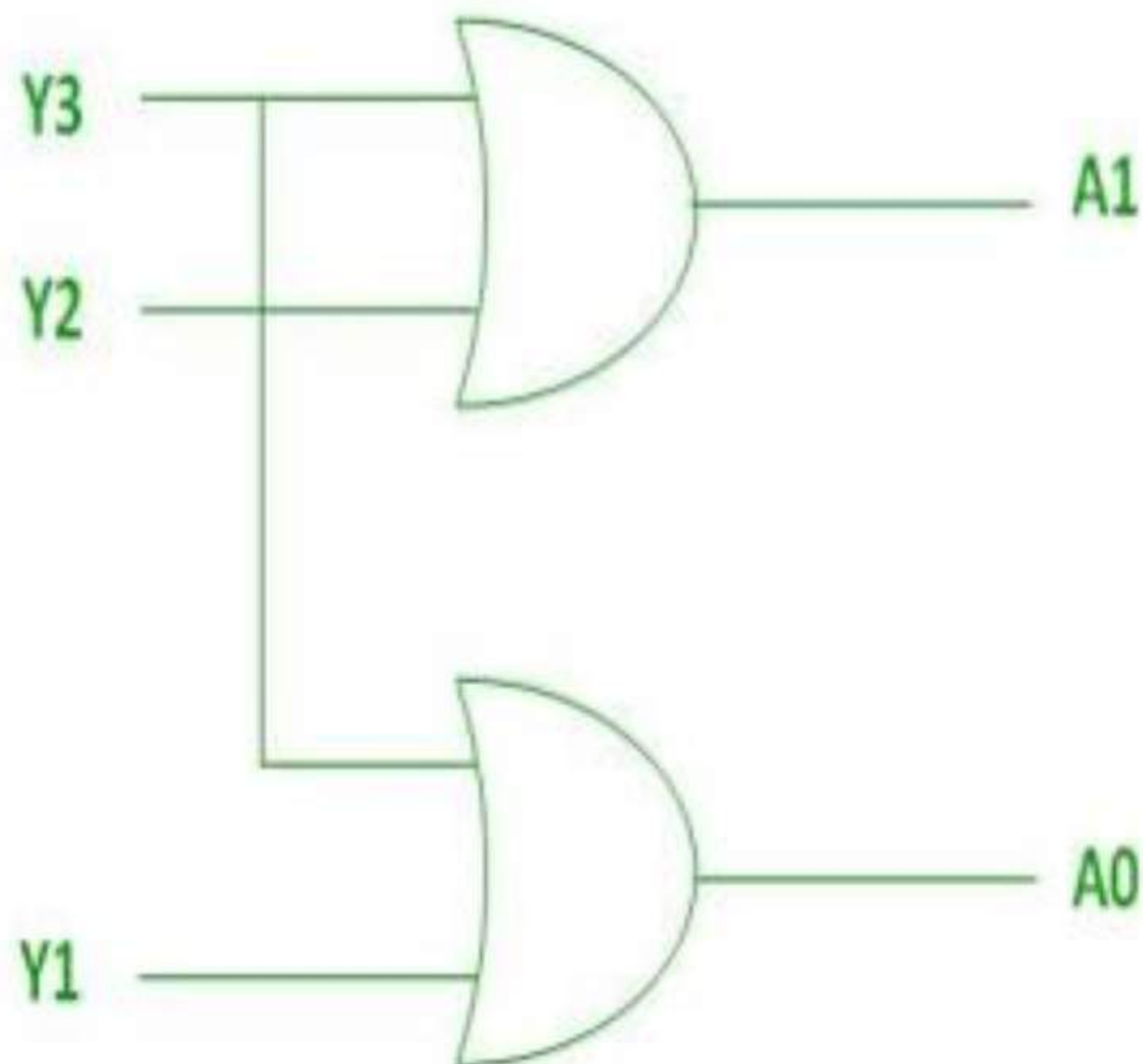
Topic :4 to 2 Encoder



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The Truth table of 4 to 2 encoders is as follows

INPUTS				OUTPUTS	
Y3	Y2	Y1	Y0	A1	A0
0	0	0	1	0	0
0	0	1	0	0	1
0	1	0	0	1	0
1	0	0	0	1	1



Logical expression for A1 and A0:

$$A1 = Y3 + Y2$$

$$A0 = Y2 + Y1$$



The 8 to 3 Encoder or octal to Binary encoder consists of **8 inputs**: Y7 to Y0 and **3 outputs**: A2, A1 & A0. Each input line corresponds to each octal digit value and three outputs generate corresponding binary code. The figure below shows the logic symbol of octal to the binary encoder.





Logical expression for A2, A1, and A0.

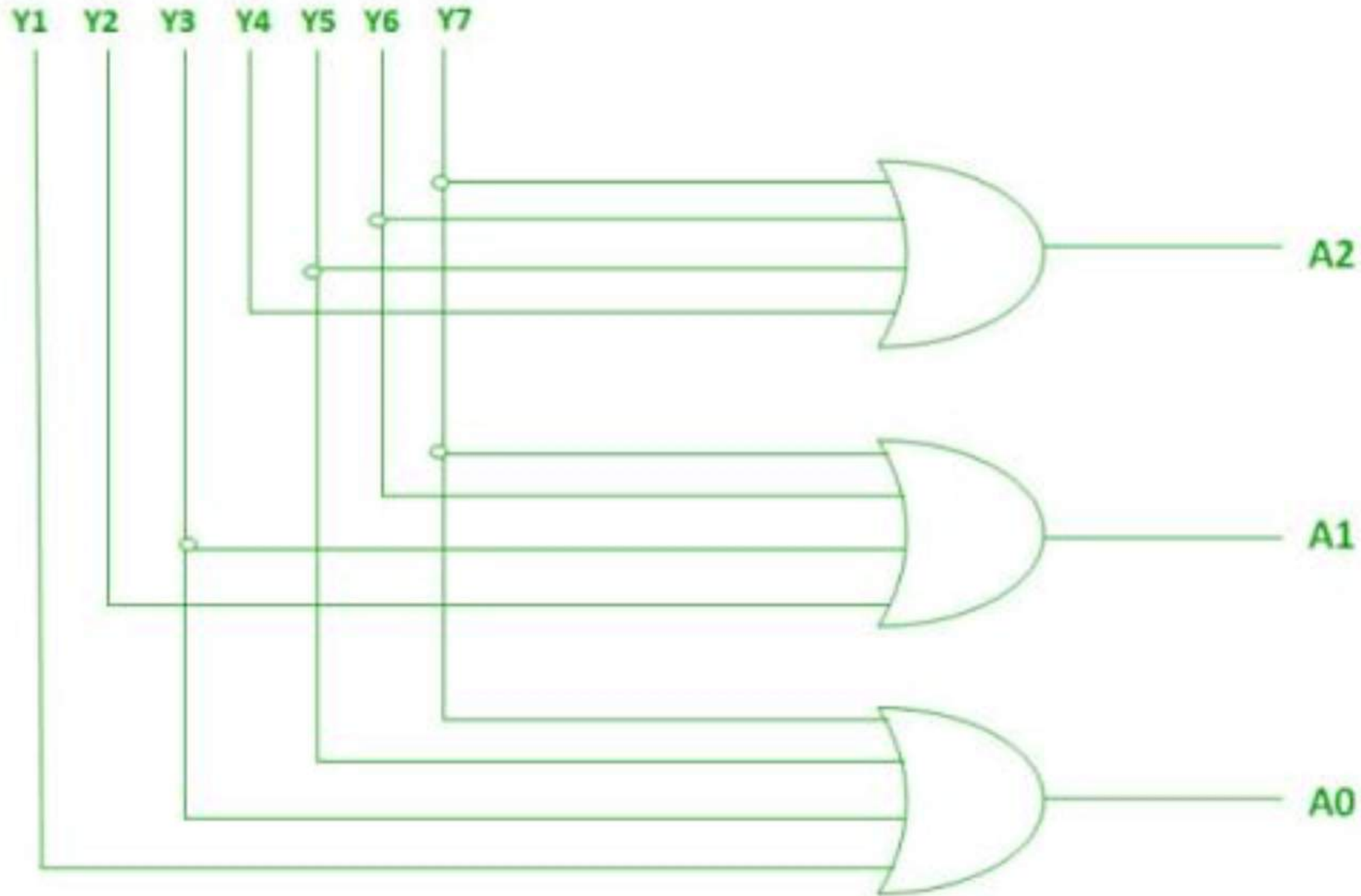
$$A_0 = Y_7 + Y_5 + Y_3 + Y_1$$



Topic : Octal to Binary Encoder



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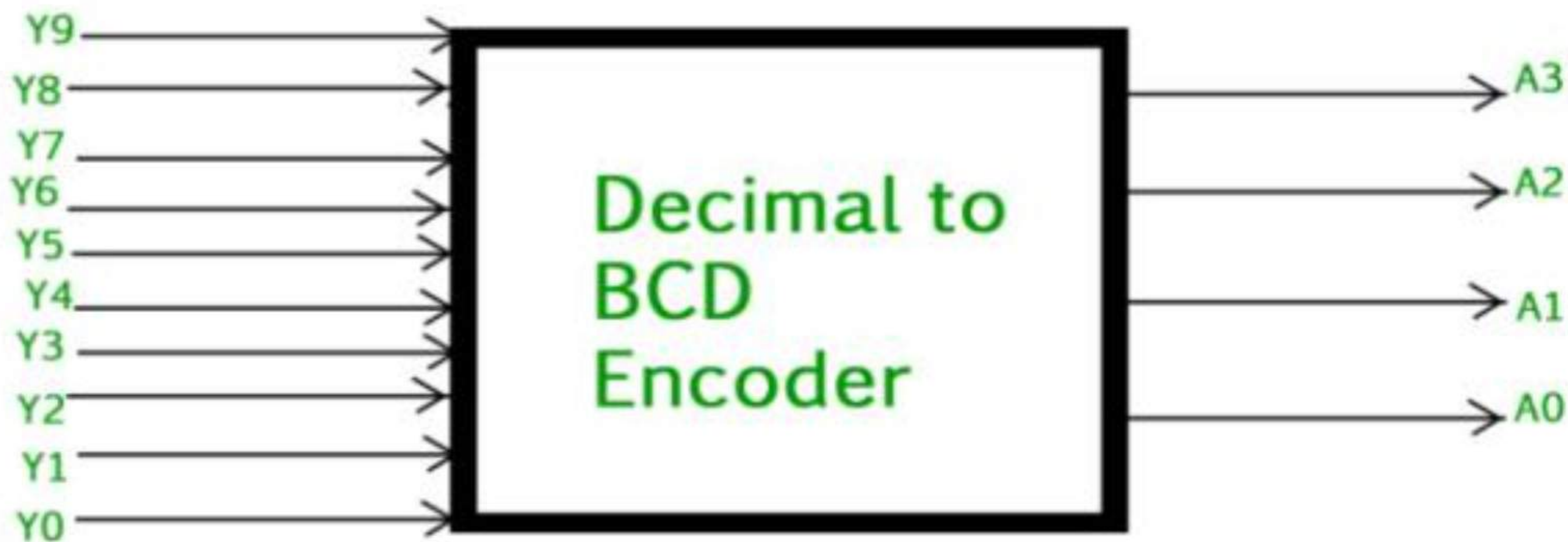


Topic : Decimal to Binary coded Encoder



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The decimal-to-binary encoder usually consists of **10 input lines** and **4 output lines**. Each input line corresponds to each decimal digit and 4 outputs correspond to the BCD code. This encoder accepts the decoded decimal data as an input and encodes it to the BCD output which is available on the output lines. The figure below shows the logic symbol of the decimal to BCD encoder :





Topic : Decimal to Binary coded Encoder



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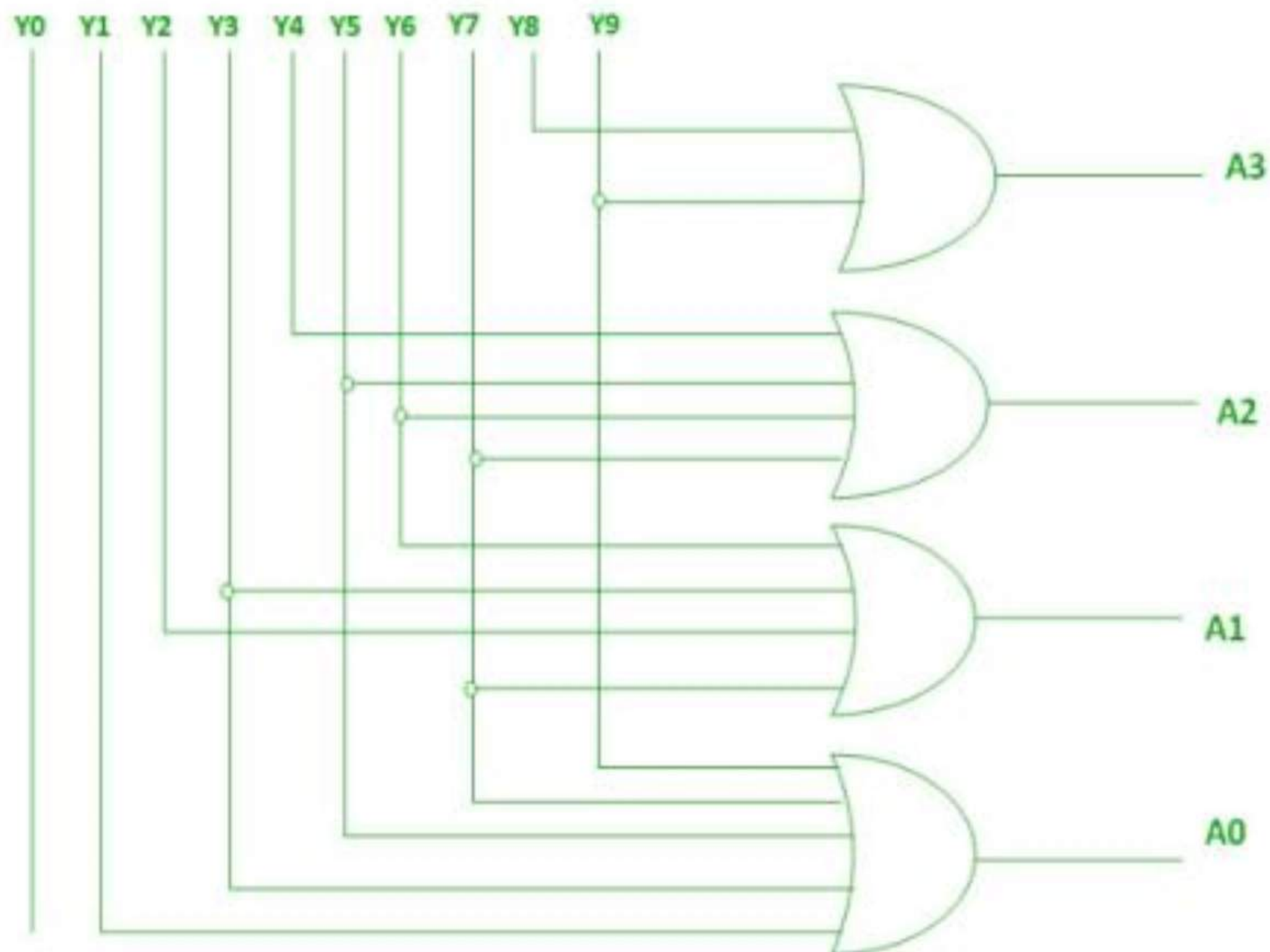
INPUTS										OUTPUTS			
Y9	Y8	Y7	Y6	Y5	Y4	Y3	Y2	Y1	Y0	A3	A2	A1	A0
0	0	0	0	0	0	0	0	0	1	0	0	0	0
0	0	0	0	0	0	0	0	1	0	0	0	0	1
0	0	0	0	0	0	0	1	0	0	0	0	1	0
0	0	0	0	0	0	1	0	0	0	0	0	1	1
0	0	0	0	0	1	0	0	0	0	0	1	0	0
0	0	0	0	1	0	0	0	0	0	0	1	0	1
0	0	0	1	0	0	0	0	0	0	0	1	1	0
0	0	1	0	0	0	0	0	0	0	0	1	1	1
0	1	0	0	0	0	0	0	0	0	1	0	0	0
1	0	0	0	0	0	0	0	0	0	1	0	0	1



Topic : Decimal to Binary coded Encoder



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Logical expression for A_3 , A_2 , A_1 , and A_0 .

$$A_3 = Y_9 + Y_8$$

$$A_2 = Y_7 + Y_6 + Y_5 + Y_4$$

$$A_1 = Y_7 + Y_6 + Y_3 + Y_2$$

$$A_0 = Y_9 + Y_7 + Y_5 + Y_3 + Y_1$$



A 4 to 2 priority encoder has **4 inputs**: Y3, Y2, Y1 & Y0, and **2 outputs**: A1 & A0. Here, the input, Y3 has the **highest priority**, whereas the input, Y0 has the **lowest priority**. In this case, even if more than one input is '1' at the same time, the output will be the (binary) code corresponding to the input, which is having **higher priority**. The truth table for the priority encoder is as follows.

INPUTS				OUTPUTS		
Y3	Y2	Y1	Y0	A1	A0	V
0	0	0	0	X	X	0
0	0	0	1	0	0	1
0	0	1	X	0	1	1
0	1	X	X	1	0	1
1	X	X	X	1	1	1



The logical expression for A1 is shown below.

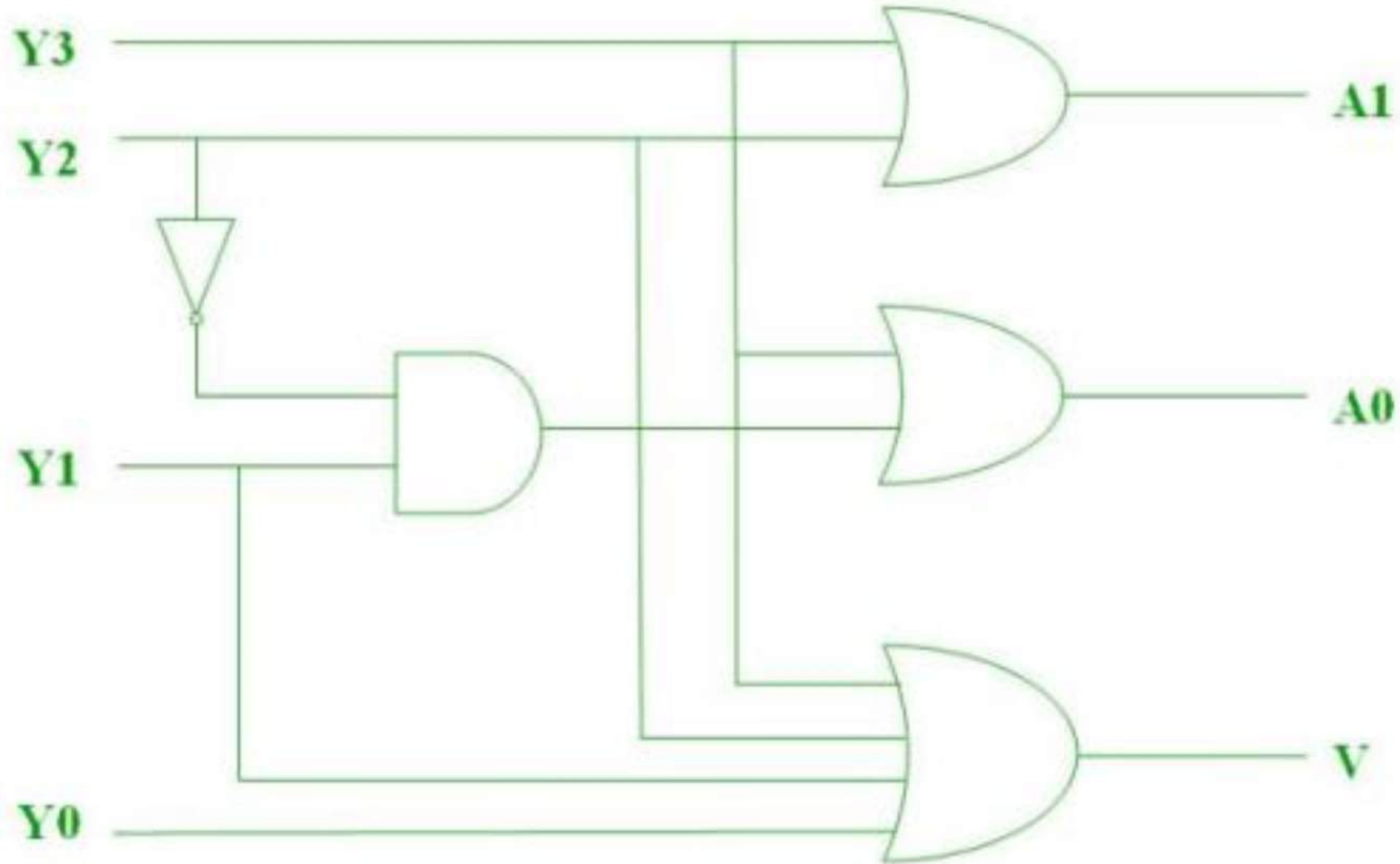
		Y1 Y0			
		00	01	11	10
Y3 Y2	00	X	0	0	0
	01	1	1	1	1
	11	1	1	1	1
	10	1	1	1	1

$$A1 = Y3 + Y2$$

The Logical Expression for A0 is shown below.

		Y1 Y0			
		00	01	11	10
Y3 Y2	00	X	0	1	1
	01	0	0	0	0
	11	X	X	X	X
	10	1	1	1	1

$$A0 = Y3 + Y2' Y1$$





Application of Encoders

- Encoders are very common electronic circuits used in all digital systems.
- Encoders are used to translate the decimal values to the binary in order to perform binary functions such as addition, subtraction, multiplication, etc.
- Other applications especially for Priority Encoders may include detecting interrupts in microprocessor applications.

Advantages of Encoders in Digital Logic

- **Reduction in the number of lines:** Encoders reduce the number of lines required to transmit information from multiple inputs to a single output, which can simplify the design of the system and reduce the cost of components.
- **Improved reliability:** By converting multiple inputs into a single serial code, encoders can reduce the possibility of errors in the transmission of information.
- **Improved performance:** Encoders can enhance the performance of a digital system by reducing the amount of time required to transmit information from multiple inputs to a single output.



Disadvantages of Encoders in Digital Logic

- **Increased complexity:** Encoders are typically more complex circuits compared to [multiplexers](#) , and require additional components to implement.
- **Limited to specific applications:** Encoders are only suitable for applications where a parallel set of inputs must be converted into a serial code.
- **Limited flexibility:** Encoders are limited in their flexibility, as they can only encode a fixed number of inputs into a fixed number of outputs.

Conclusion

Here we have learned about encoder on digital logic that ,an encoder is a device in digital electronics that takes multiple input signals and turns them into a smaller, coded output. Encoders are used in things like keyboards or communication systems to send or store information in a more compact way. We have learned about types of encoder, advantages disadvantages and applications of it. This helps make systems more efficient and less complex.

THANK YOU